

LAUGHLIN BULLHEAD, AZ, USA

WMO: 723788

Lat: 35.157N

Lon: 114.559W

Elev: 695

StdP: 14.33

Time Zone: -7.00 (W07)

Period: 97-19

WBAN: 53135

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	37.1	40.8	0.6	5.8	61.6	4.9	7.3	63.0	36.3	51.8	33.8	53.5	5.8	30	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	112.9	68.6	110.6	68.2	108.5	67.9	74.4	95.0	73.5	95.1	72.5	95.8	10.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	111.7	81.1	66.3	99.5	84.0	63.9	91.4	87.2	38.3	93.8	37.2	94.1	36.4	97.0	93.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.8	26.8	24.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.7	55.9	59.3	66.3	72.5	80.0	91.2	95.9	94.5	87.7	75.4	63.1	54.1		
	DBStd	15.80	5.80	6.52	6.42	6.74	6.96	6.01	4.38	4.93	5.63	6.71	6.62	6.10		
	HDD50	54	18	5	0	0	0	0	0	0	0	0	2	29		
	HDD65	1000	284	177	64	17	1	0	0	0	0	5	112	340		
	CDD50	9089	201	265	506	674	932	1237	1423	1380	1132	789	395	155		
	CDD65	4557	2	17	104	240	467	787	958	915	682	329	55	1		
	CDH74	70883	5	174	1179	2904	6495	13402	16862	15669	10206	3518	464	5		
(13)	CDH80	47005	0	22	405	1346	3723	9399	12450	11349	6565	1637	109	0		
(14)	Wind	WSAvg	9.2	12.2	9.7	9.2	9.5	8.7	8.1	7.2	7.0	8.7	9.5	11.3		
(15) Precipitation	PrecAvg	5.0	0.8	1.0	0.6	0.2	0.0	0.0	0.2	0.3	0.3	0.3	0.3	0.4		
	PrecMax	11.0	4.3	4.7	4.4	0.7	0.6	0.4	1.4	1.1	1.6	2.2	1.3	1.9		
	PrecMin	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.9	1.2	1.1	1.0	0.2	0.1	0.1	0.3	0.3	0.4	0.6	0.4	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	82.3	92.8	99.5	107.5	117.9	115.4	113.4	109.1	99.6	86.3	73.4		
		MCWB	51.3	54.6	58.8	60.4	63.7	68.1	69.3	69.8	68.0	64.6	58.1	54.6		
	2%	DB	71.9	79.2	88.4	95.0	103.7	112.7	112.9	111.2	106.2	95.0	83.7	71.5		
		MCWB	50.1	53.4	57.3	58.8	62.2	66.1	69.4	69.3	68.1	62.1	57.3	52.6		
	5%	DB	68.2	75.3	84.2	91.1	99.9	109.1	110.5	109.1	103.9	91.4	80.9	67.5		
		MCWB	48.9	52.3	55.4	57.4	60.8	65.9	69.2	68.7	67.3	60.9	55.6	50.0		
	10%	DB	65.9	72.2	81.2	88.2	96.6	107.5	108.3	108.0	101.7	89.8	75.5	64.1		
		MCWB	48.5	50.9	54.1	56.4	59.7	65.0	68.5	68.6	66.6	61.1	53.5	48.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.8	59.5	59.7	62.3	64.3	72.1	74.9	75.8	76.0	68.1	60.2	58.6		
		MCDB	63.5	73.4	86.2	92.8	99.9	103.8	95.4	95.4	90.8	84.4	79.6	67.2		
	2%	WB	54.0	56.2	58.2	60.3	62.6	69.8	73.8	74.8	73.7	66.2	58.2	55.7		
		MCDB	63.6	70.1	82.8	88.2	96.6	104.8	96.3	95.8	90.1	84.9	79.4	64.9		
	5%	WB	52.4	53.9	56.8	59.0	61.6	67.9	73.1	73.8	72.2	64.7	56.5	53.4		
		MCDB	63.2	70.7	80.6	85.8	94.3	104.9	97.0	95.4	91.4	84.0	77.5	63.7		
	10%	WB	50.4	52.4	55.1	57.4	60.2	66.2	72.2	72.8	70.8	63.0	54.8	50.8		
		MCDB	61.7	70.1	78.8	84.9	92.1	103.6	97.8	95.5	91.9	82.5	75.1	60.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	19.9	23.2	23.9	23.5	25.5	22.5	23.1	23.7	21.7	20.0	15.8		
		MCDBR	20.5	25.9	29.4	29.4	29.1	28.2	24.9	25.3	26.7	24.8	24.6	19.8		
		MCWBR	9.8	11.5	11.7	11.5	10.5	9.3	7.9	8.3	9.1	9.3	10.6	9.4		
	5% WB	MCDBR	16.2	23.1	26.8	26.1	26.5	25.5	19.5	20.0	20.4	20.1	22.1	16.5		
		MCWBR	8.1	10.5	10.7	9.9	9.8	8.5	6.0	6.7	8.1	7.2	10.0	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.284	0.298	0.337	0.349	0.348	0.399	0.396	0.350	0.311	0.284	0.278		
	taud		2.579	2.583	2.523	2.390	2.387	2.392	2.362	2.364	2.457	2.542	2.586	2.593		
	Ebn at Noon		298	308	311	300	296	295	279	279	288	293	294	292		
	Edh at Noon		24	27	31	37	38	38	39	38	33	28	24	23		
(44) All-Sky Solar Radiation	RadAvg		1021	1330	1802	2247	2546	2718	2395	2221	1972	1512	1138	907		
	RadStd		72	105	84	86	95	92	114	102	75	78	47	64		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	-3.08	N/A	-0.72	N/A	N/A	N/A	+299	+140

Nomenclature: See separate page